

Ten Popular Tools and Frameworks for Artificial Intelligence

This article highlights ten tools and frameworks that feature on the 'hot list' for artificial intelligence. A short description along with features and links is given for each tool or framework.



Let's go on an exciting journey, discovering exactly why the following tools and frameworks are ranked so high.

1) TensorFlow: An open source software library for machine intelligence

TensorFlow is an open source software library that was originally developed by researchers and engineers working on the Google



Brain Team. TensorFlow is used for numerical computation with data flow graphs. Nodes in the graph represent mathematical operations, while the graph edges represent the multi-dimensional data arrays (tensors) communicating between them. The flexible architecture allows you to deploy computation to one or more

CPUs or GPUs in a desktop, server or mobile device, with a single API.

TensorFlow provides multiple APIs. The lowest level API—TensorFlow Core—provides you with complete programming control. The higher-level APIs are built on top of TensorFlow Core and are typically easier to learn and use than TensorFlow Core. In addition, the higher-level APIs make repetitive tasks easier and more consistent between different users. A high-level API like *tf.estimator* helps you manage data sets, estimators, training and inference.

The central unit of data in TensorFlow is the tensor, which consists of a set of primitive values shaped into an array of any number of dimensions. A tensor's rank is its number of dimensions.

A few Google applications using TensorFlow are listed below.

RankBrain: A large-scale deployment of deep neural nets for search ranking on www.google.com.

Inception image classification model: This is a baseline model, the result of ongoing research into highly accurate computer vision models, starting with the model that won the 2014 Imagenet image classification challenge.

SmartReply: A deep LSTM model to automatically generate email responses.

Massive multi-task networks for drug discovery: A deep neural network model for identifying promising drug candidates – built by Google in association with Stanford University.

On-device computer vision for OCR: An on-device computer vision model for optical character recognition to enable real-time translation.

Useful links

Tensorflow home: <https://www.tensorflow.org>

GitHub: <https://github.com/tensorflow>

Getting started: https://www.tensorflow.org/get_started/get_started